Strength of materials Lab. Manual

Production Engineering

Strength of materials lab. manual

Contents

S.No.	Title	Pg.no
1.	Rockwell Hardness test	3
2.	Brinell hardness test.	5
3.	Impact test	8
4.	Tension test	14
5.	Torsion test	19
6.	Bending test	21
7.	Shear test	24
8.	Compression test	26

Instructions.-1.Write observations, tables, diagrams, Specimen calculations in the blank left side of the journal and others to the right.

Experiment No.1

Title **Rockwell Hardness test**

Objective To determine the hardness the Hardness of the given Specimen using Rockwell hardness test.

Materials and equipments required

Rockwell hardness testing machine.
Black diamond cone indenter,
Hard steel specimen.

Theory

Rockwell test is developed by the Wilson instrument co U.S.A in 1920.

This test is an indentation test used for smaller specimens and harder materials. The test is subject of IS: 1586.In this test indenter is forced into the surface of a test piece in two operations, measuring the permanent increase in depth of an indentation from the depth increased from the depth reached under a datum load due to an additional load.

Measurement of indentation is made after removing the additional load. Indenter used is the cone having an angle of 120 degrees made of black diamond.

Precautions

1. Thickness of the specimen should not be less than 8 times the depth of indentation to avoid the deformation to be extended to the opposite surface of a specimen.
2. Indentation should not be made nearer to the edge of a specimen to avoid unnecessary concentration of stresses. In such case distance from the edge to the center of indentation should be greater than 2.5 times diameter of indentation.
3. Rapid rate of applying load should be avoided. Load applied on the ball may rise a little because of its sudden action. Also rapidly applied load will restrict plastic flow of a material, which produces effect on size of indentation.

Procedure

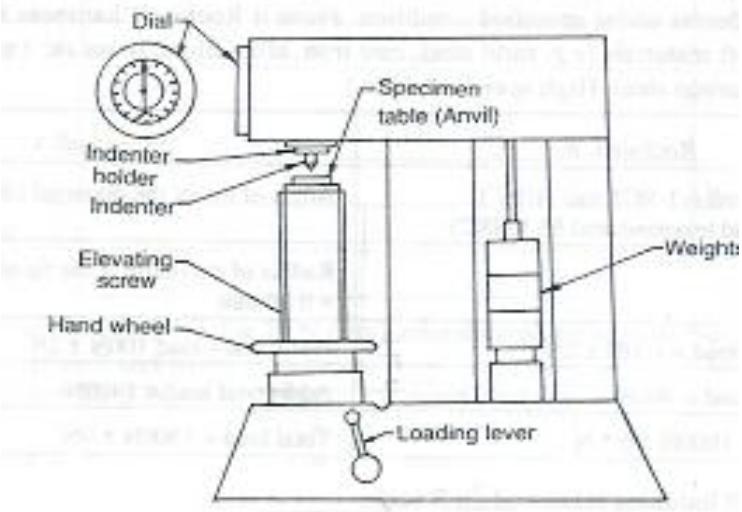
1. Examine hardness testing machine (fig.1).
2. Place the specimen on platform of a machine. Using the elevating screw raise the platform and bring the specimen just in contact with the ball. apply an initial load until the small pointer shows red mark.
3. Release the operating valve to apply additional load. Immediately after the additional load applied, bring back operating valve to its position.
4. Read the position of the pointer on the C scale, which gives the hardness number.

5. Repeat the procedure five times on the specimen selecting different points for indentation.

Observation

1. Take average of five values of indentation of each specimen. Obtain the hardness number from the dial of a machine.
2. Compare Brinell and Rockwell hardness tests obtained.

Figure .1



Rockwell hardness test equipment

Result

Rockwell hardness of given specimen is ---------

\

Experiment No.2

Title **Brinell hardness test.**

Aim To determine the hardness of the given specimen using Brinell hardness test.

Specimen and specimen

Brinell hardness tester (fig.2)
Aluminum specimen
Ball indenter.

Precautions

1. Thickness of the specimen should not be less than 8 times the depth of indentation to avoid the deformation to be extended to the opposite surface of a specimen.
2. Indentation should not be made nearer to the edge of a specimen to avoid unnecessary concentration of stresses. In such case distance from the edge to the center of indentation should be greater than 2.5 times diameter of indentation.
3. Rapid rate of applying load should be avoided. Load applied on the ball may rise a little because of its sudden action. Also rapidly applied load will restrict plastic flow of a material, which produces effect on size of indentation.
4. Surface of the specimen is well polished, free from oxide scale and any foreign material.

Theory

Hardness of a material is generally defined as Resistance to the permanent indentation under static and dynamic load. When a material is required to use under direct static or dynamic loads, only indentation hardness test will be useful to find out resistance to indentation.

In Brinell hardness test, a steel ball of diameter (D) is forced under a load (F) on to a surface of test specimen. Mean diameter (d) of indentation is measured after the removal of the load (F).

Observation

1. Take average of five values of indentation of each specimen. Obtain the hardness number from equation (!).
2. Compare Brinell and Rockwell hardness tests obtained.

Procedure

1.Load to be applied for hardness test should be selected according to the expected hardness of the material. However test load shall be kept equal to 30 times the square of the diameter of the ball (diameter in mm)

$F=30.D^2$

Where ball diameter, generally taken as 10 mm.

For guidelines hardness range for standard loads given below

Ball diameter	Load (kg)	Range of Brinell hardness
10	3000	96 to 600
	1500	48 to 300
	500	16 to 100

2.Apply the load for a minimum of 15 seconds to 30 seconds. [if ferrous metals are to be tested time applied will be 15 seconds and for softer metal 30 seconds]

3.Remove the load and measure the diameter of indentation nearest to 0.02 mm using microscope (projected image)

4.Calculate Brinell hardness number (HB). As per IS: 1500.

5.Brinell hardness number

$$\frac{2F}{\pi D\left[D-\sqrt{D^2-d^2}\right]} \qquad (1)$$

where D is the diameter of ball indenter and d is the diameter of indentation.
Hardness numbers normally obtained for different materials are given below (under 3000 kg and 10 mm diameter ball used)

Ordinary steels medium carbon	100 to 500
Structural steel	130 to 160
Very hard steel	800 to 900

Note: Brinell test is not recommended for then materials having HB over 630.
It is necessary to mention ball size and load with the hardness test when standard size of ball and load are not used. Because indentation done by different size of ball and load on different materials are not geometrically similar. Ball also

undergoes deformation when load is applied. Material response to the load is not same all the time.

6.Brinell hardness numbers can be obtained from tables 1 to 5 given in IS: 1500, knowing diameter of indentation, diameter of the ball and load applied.

Figure 2

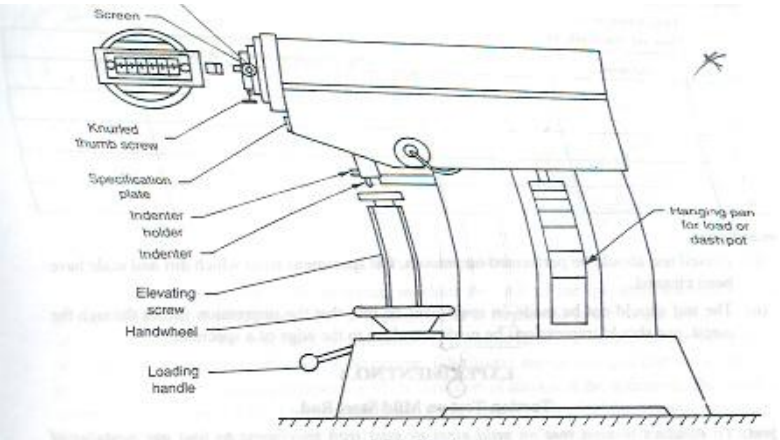


Fig. 7. Brinell hardness tester.

Brinell hardness number, $BHN = \frac{2P}{\pi D\,(D - \sqrt{D^2 - d^2})}$.

Test Requirements/Procedure:

Brinell hardness tester

Result;

The Brinell hardness number of the specimen is -------

Experiment No.3

Title **Impact test**

Aim To determine the Impact toughness (strain energy) through Izod test and Charpy test

Theory

In a impact test a specially prepared notched specimen is fractured by a single blow from a heavy hammer and energy required being a measure of resistance to impact.

Impact load is produced by a swinging of an impact weight W (hammer) from a height h. Release of the weight from the height h swings the weight through the arc of a circle, which strikes the specimen to fracture at the notch (fig..

Kinetic energy of the hammer at the time of impact is $mv^2/2$, which is equal to the relative potential energy of the hammer before its release. (mgh),where m is the mass of the hammer and $v = \sqrt{2gh}$ is its tangential velocity at impact, g is gravitational acceleration (9.806 m/s^2) and h is the height through which hammer falls. Impact velocity will be 5.126 m/s or slightly less.
Here it is interesting to note that height through which hammer drops determines the velocity and height and mass of a hammer combined determine the energy.

Energy used can be measured from the scale given. The difference between potential energies is the fracture energy. In test machine this value indicated by the pointer on the scale. If the scale is calibrated in energy units, marks on the scale should be drawn keeping in view angle of fall () and angle of rise (. Height h_1 and h_2 equals,

h_1= R (1-cos θ) and h_2= (1-cos θ).

With the increase or decrease in values, gap between marks on scale showing energy also increase or decrease. This can be seen from the attached scale with any impact machine.

Energy used in fracturing the specimen can be obtained approximately as
Wh_1-Wh_2

This energy value called impact toughness or impact value, which will be measured, per unit area at the notch.

Izod introduced Izod test in 1903. Test is as per the IS: 1598
Charpy introduced Charpy test in 1909. Test is as per the IS: 1499.

a. Izod test

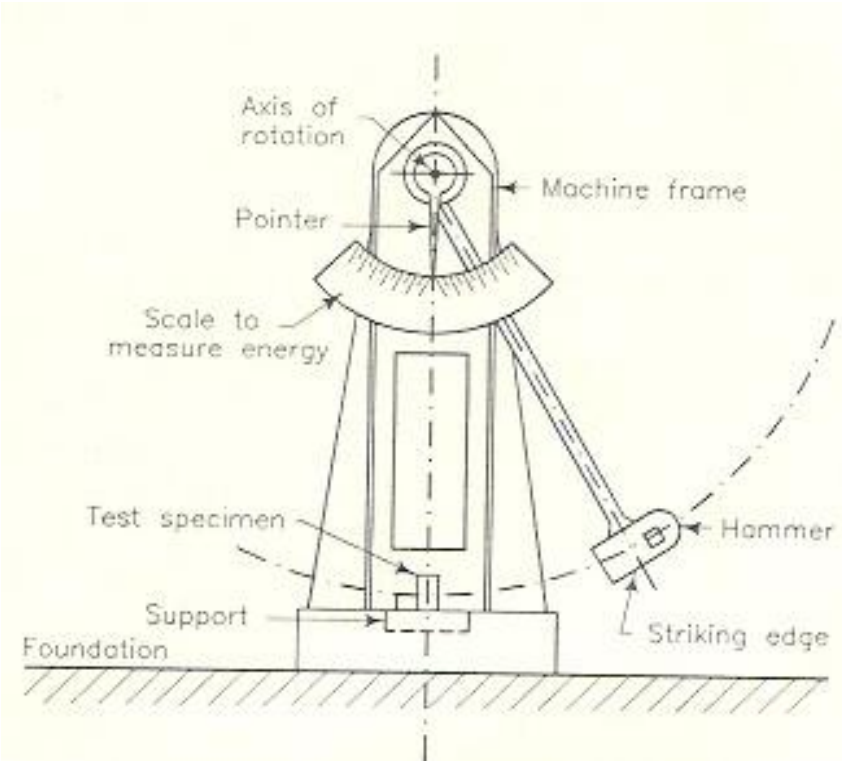
Specimen and equipment

1. Impact testing machine.(fig.3)
2. Specimen and v notch is shown in the fig.4. Size of the specimen is 10mm X 10mm X 75mm

Mounting of the specimen:

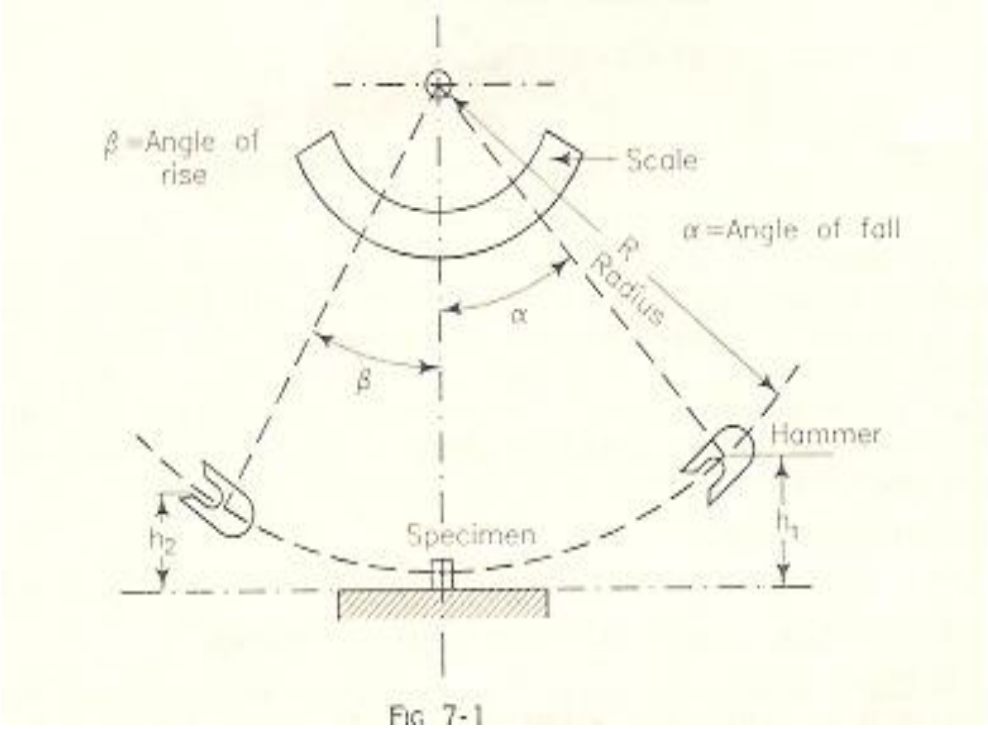
Specimen is clamped to act as vertical cantilever with the notch on tension side. Direction of blow of hammer is shown in fig. (). Direction of blow is shown in fig

Figure. **3.a**



Izod Impact testing equipment

Figure 3.b



Schematic impact testing

Figure 4

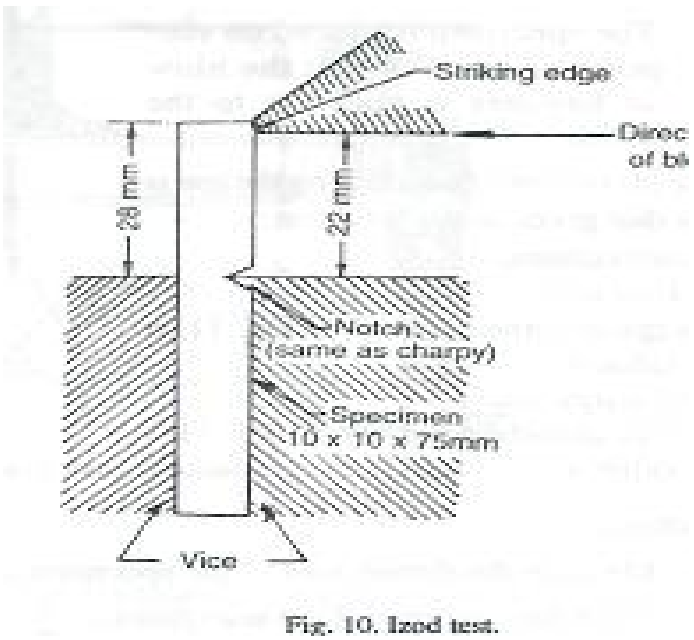


Fig. 10. Izod test.

Position of specimen for Izod test

Procedure

1. Measure the dimensions of a specimen. Also, measure the dimensions of The notch.

2. Raise the hammer and note down initial reading from the dial, which will be energy to be used to fracture the specimen.

3. Place the specimen for test and see that it is placed center with respect to hammer. Check the position of notch.

4. Release the hammer and note the final reading. Difference between the initial and final reading will give the actual energy required to fracture the Specimen.

5. Repeat the test for specimens of other materials.

6. Compute the energy of rupture of each specimen.

Observation

Initial and final reading of the dial.

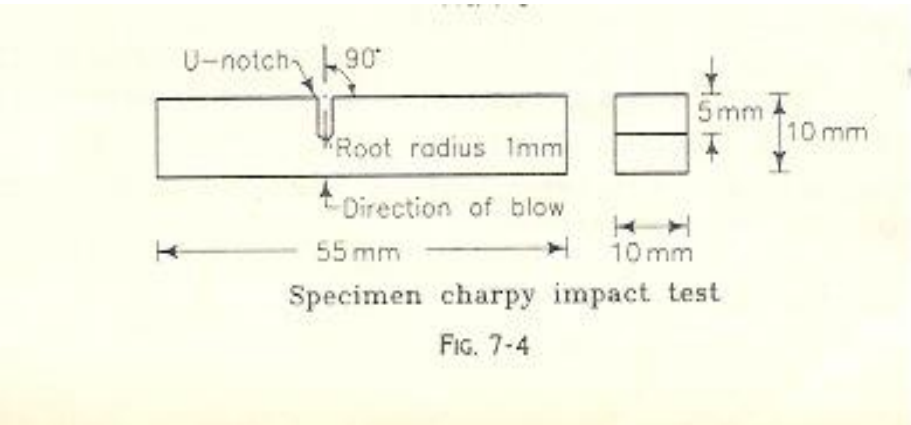
Result Strain energy of given specimen is ---------

b. Charpy test

Specimen and equipment:

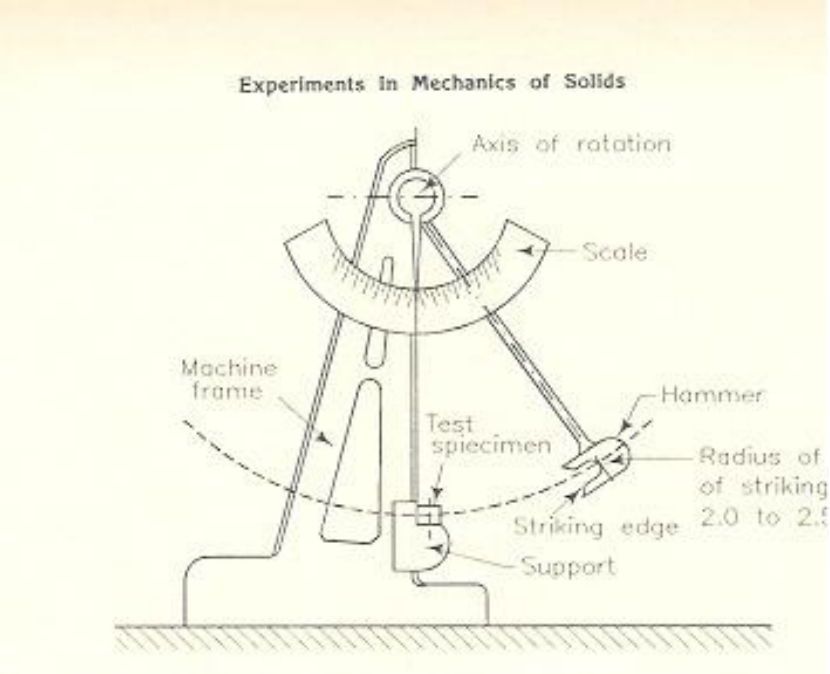
1. Impact testing machine. (Fig.6)
2. U notch is cut across the middle of one face as shown in (fig.5).

Figure 5



Specimen for Charpy test

Figure 6

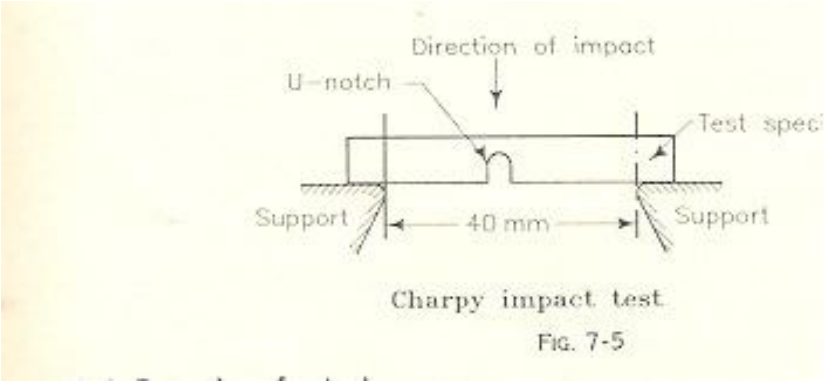


Charpy impact testing equipment

Mounting of specimen

Specimen is tested as a beam supported at each end (fig.7). Hammer is allowed to hit then specimen at the opposite face behind the notch.

Figure.7



Mounting of specimen

Procedure

1. Measure the dimensions of a specimen. Also, measure the dimensions of The notch.

2. Raise the hammer and note down initial reading from the dial, which will be energy to be used to fracture the specimen.

3. Place the specimen for test and see that it is placed center with respect to hammer. Check the position of notch.

4. Release the hammer and note the final reading. Difference between the initial and final reading will give the actual energy required to fracture the Specimen.

5. Repeat the test for specimens of other materials.

6. Compute the energy of rupture of each specimen.

Observation

Initial and final reading of the dial.

Result Strain energy of given specimen is ---------

Experiment No.4

Title: **Tension test**

Aim: To determine the tensile strength of specimen

Specimen and equipments

Universal testing machine (fig7.a)
Specimen as shown in the(fig7.b)
Of different ferrous and non ferrous materials

Figure 7.a

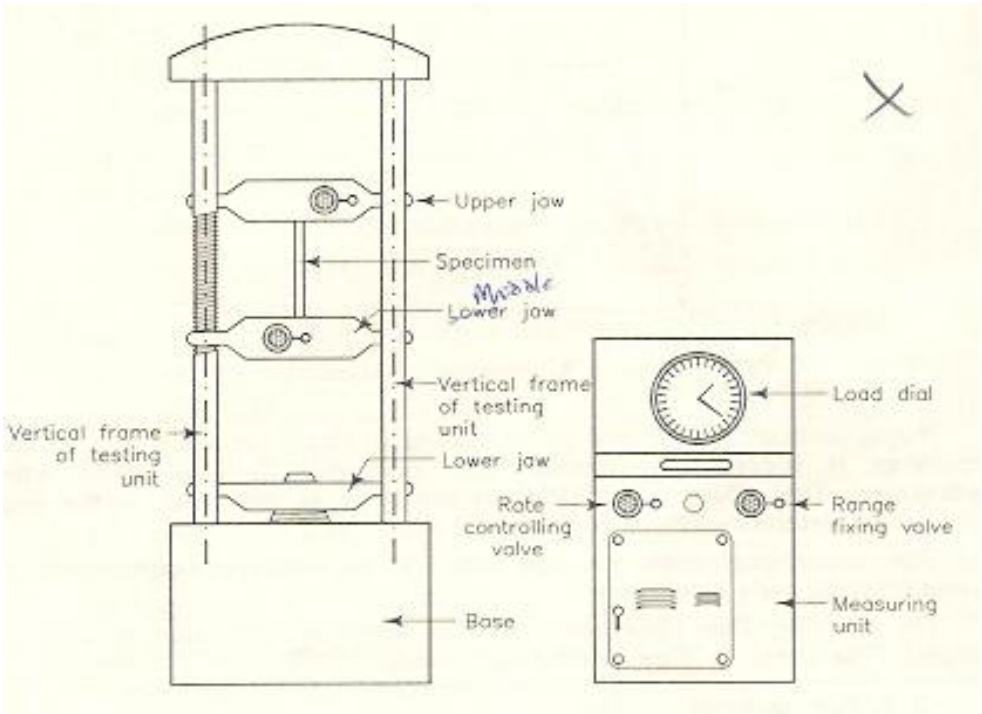


Figure.7b

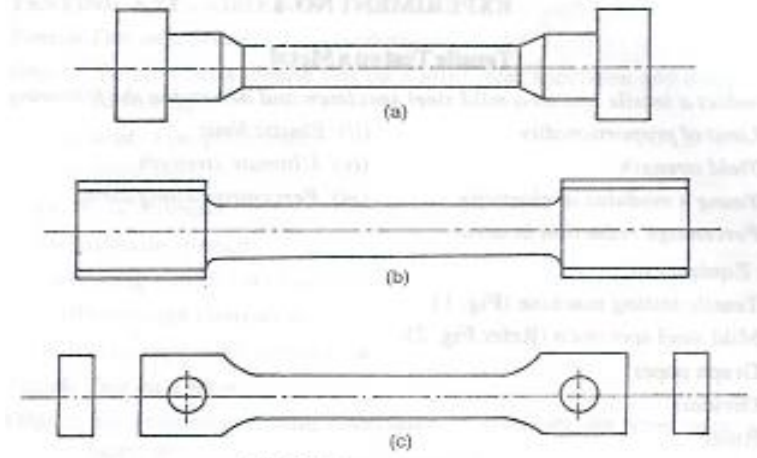


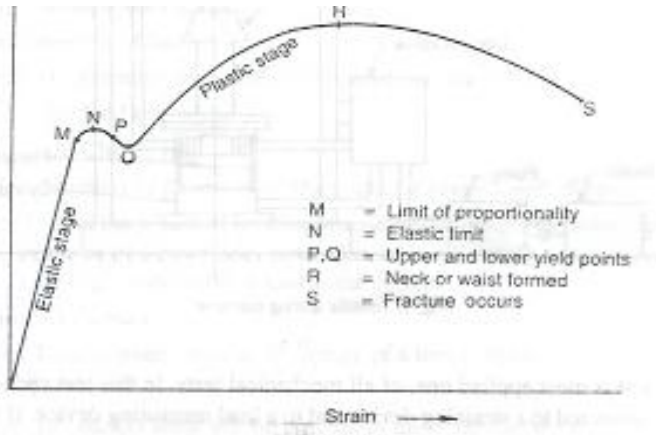
Fig. 2. Mild steel specimens.

Theory

The tensile test is most applied one, of all mechanical tests. In this test ends of a test piece are fixed into grips connected to a straining device and to a load-measuring device. If the applied load is small enough, the deformation of any solid body is entirely elastic. An elastically deformed solid will return to its original position as soon as load is removed. However, if the load is too large, the material can be deformed permanently. The initial part of the tension curve (fig.8), which is recoverable immediately after unloading, is termed as elastic and rest of the curve, which represents the manner in which solid undergoes plastic deformation is termed plastic. the stress below which the deformation is essentially entirely elastic is known as the yield strength of material. In some materials (like mild steel) the onset of plastic deformation is denoted by a sudden drop in load indicating both an upper and lower yield point. However, some materials do not exhibit a sharp yield point. During plastic deformation, at larger extensions strain hardening cannot compensate for the decrease in section and thus the load passes trough a maximum and then begins to decrease. As this stage the' Ultimate strength ', which is defined as the ratio of the specimen to original cross –sectional area, reaches a maximum value. Further loading will eventually cause 'neck' formation and rupture.

Usually a tension test is conducted at room temperature and the tensile load is applied slowly. During this test either round or flat specimens (fig.7) may be used. The round specimens may have smooth, shouldered or threaded ends. The load on the specimen is applied mechanically or hydraulically depending on the type of testing machine.

Figure. 8



Stress-strain diagram

Procedure

1. Measure the dimensions of a specimen

   Diameter=d= -----,
   Total length of a specimen,
   Cross sectional area = Ao= ----,

   Mark gage length (L_o) at three different portions on the specimen, covering effective length of a specimen.(this is required so that necked portion will remain between any two points of gage length on the specimen.)

2. Grip the specimen in the fixed head of a machine. (Portion of the specimen has to be gripped as shown in the fig.7.

3. Fix the extensometer within the gauge length marked on the specimen. Adjust the dial of extensometer at zero.

4. Adjust the dial of a machine to zero, to read load applied.

5. Select suitable increments of loads to be applied so that corresponding elongation can be measured from dial gauge.

6. Keep speed of machine uniform. Record yield point, maximum load point, point of breaking of specimen.

7. Remove the specimen from machine and study the fracture observes type of fracture.

8. Measure dimensions of tested specimen. Fit the broken parts together and measure reduced diameter and final gage length.

Observations

Specimen prepared from M.S bar/CI/Al

1. Diameter = d = ---- mm

2. Gage length (l_o)= 5Xd= -----mm

3. Original cross sectional area of the specimen
= A_o = ------mm^2

4. Final gage length obtained= Lo'=

5. Final diameter obtained = ----mm

Observation table 1

Sr. No	Load applied (N) (p)	Area of a specimen (A_o)	Stress N/mm^2	Modulus of elasticity (E) N/mm^2

Observation table 2.

Sr. No	Contraction in diameter (d_d) (mm)	Deformation in length (mm)	Lateral strain	Linear strain	Poisson ratio

Note

1. Use vernier caliper to measure diameter, gage length etc. for the specimen.
2. If C.I. specimen is to be tested only one observation will be taken at failure.

Results

1. Calculate stress and strain for every interval of applied load.
Draw stress-strain curve as shown in the Fig.()

2. Compute the following;

a. Modulus of elasticity

Hook's law states that stress is always proportional to strain within elastic limit. The ratio of stress and strain is constant, called modulus of elasticity or young's modulus (E)

E= Stress/strain =Constant=E= -----,

b. Yield stress (f_y);

The point, at which strain increases without increase in stress, is known as Yield point. Stress measured at yield point is called yield stress.

c. **Tensile strength**:
Maximum carrying capacity of a material in tension is called tensile

strength

Tensile strength= maximum tensile load/ original cross sectional Area.

d. **Percentage elongation**:

The extension produced in a gage length, expressed as a percentage of its original value(L_O)

% Elongation=$[(L_O' - L_o)/L_o] \times 100$

where L_o' is final gage length after fracture.

e. Percentage reduction in area:

$= [(A_o - A_o')/A_o] \times 100$

where A_o' is final reduced cross sectional area after fracture.

Experiment No.5

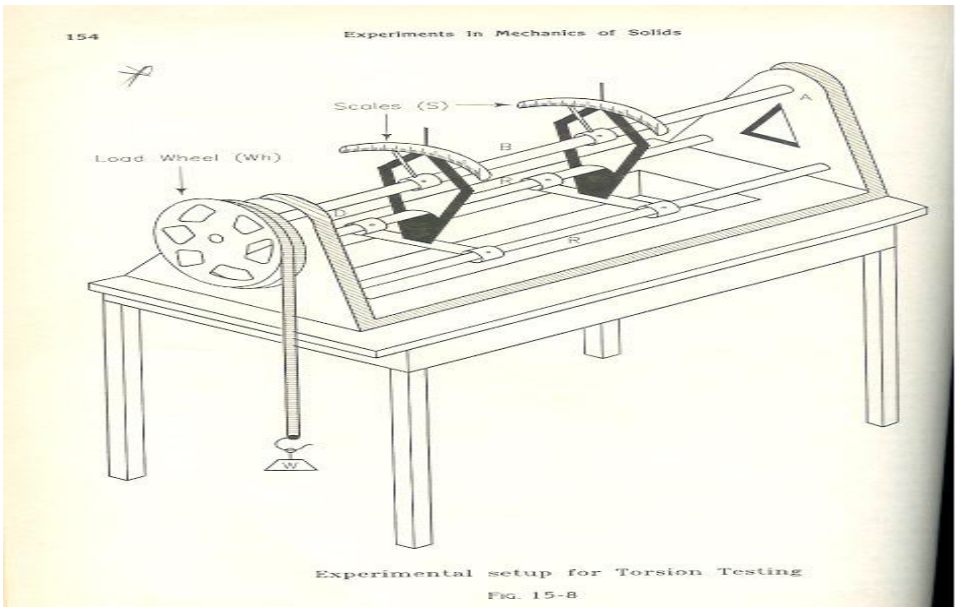
Title Torsion test

Aim: To find the modulus of rigidity.

Specimen and equipments

1. A torsion testing apparatus,
2. Standard specimen of mild steel or cast iron.
3. Twist meter for measuring angles of twist
4. A steel rule and calipers and micrometer.

Figure.9.



Torsion equipment

Theory

A torsion test is quite instrumental in determining the value of rigidity (ratio of shear stress to shear strain) of a metallic specimen. The value of modulus of rigidity can be found out through observations made during the experiment by using the torsion equation:

$$\frac{T}{I_p} = \frac{C\theta}{l} \quad \text{or} \quad C = \frac{Tl}{I\,\theta}$$

Where T=torque applied,
I_p= polar moment of inertia,
C=modulus of rigidity,
= Angle of twist (radians), and
l= gauge length.

In the torque equipment refer fig. One end of the specimen is held by a fixed support and the other end to a pulley. The pulley provides the necessary torque to twist the rod by addition of weights (w). The twist meter attached to the rod gives the angle of twist.

Procedure

1. Prepare the testing machine by fixing the two twist meters at some constant lengths from fixed support.

2. Measure the diameter of the pulley and the diameter of the rod.

3. Add weights in the hanger stepwise to get a notable angle of twist for T_1 and T_2
4. Using the above formula calculate C

Conclusion:

Result

Modulus of rigidity of the shaft -----

Experiment No.6

Title Bending test

Aim To find the values of bending stresses and young's modulus of the material of a beam (say a wooden or steel) simply supported at the ends and carrying a concentrated load at the center.

Material and equipment

1. Universal testing machine
2. Beam of different cross sections and materials (say wood or steel)

Figure.10

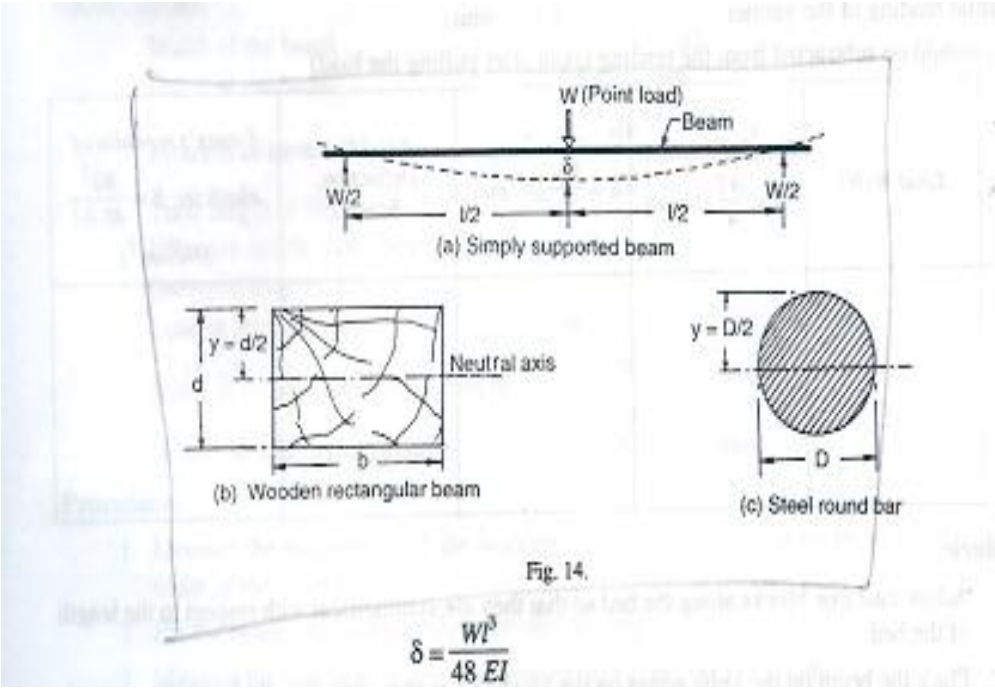


Fig. 14.

$$\delta = \frac{Wl^3}{48\,EI}$$

Specimen details and mounting

Theory

If a beam is simply supported at the ends and carries a concentrated load at the center, the beam bends concave upwards. The distance between the original position of the beam and its position after bending is different at different points (fig) along the length if the beam, being maximum at the center in this case. This difference is called 'deflection'.

In this type of loading the maximum amount of deflection () is given by the relation,

$$\delta = \frac{Wl^3}{48Ei}$$

or

$$E = \frac{Wl^3}{48EI}$$

Where W= load acting at the center, N
l=length of the beam between the supports, mm
E=young's modulus of material of the beam, N/mm^2
I=second moment of area of the cross section (moment of inertia) of the beam, about the neutral axis, mm^4

Bending stress:

As per bending equation,

$$\frac{M}{I} = \frac{\sigma_b}{y}$$

Where M= bending moment, Nmm

I= moment of inertia, mm^4
σ_b =Bending stress, N/mm^2
y=distance of the fiber of the beam from the neutral axis.

Observation

Refer Fig.

Width of the beam=……mm (for rectangular cross section)
Depth of the beam D=…mm (for circular cross section)
Moment of inertia of rectangular section= bd^3/12=……mm^4
Moment of inertia of circular section =……mm^4
Initial reading of the vernier= ….mm
(It should be subtracted from the reading taken after putting the load)

S. No	Load W(N)	Bending moment $M = \frac{Wl}{4}(N-mm^3)$	Bending stress $\sigma_b = \frac{My}{I}(N/mm^2)$	Deflection $\delta(mm)$	Young's modulus of elasticity $E = \frac{Wl^3}{48\delta I}$ (N/mm^2)

Precautions

1. Make sure that the beam and load is placed at the proper position.
2. Cross section of the beam should be large
3. Note down the readings of the vernier scale carefully.

Procedure

1. Adjust the supports alone the UTM bed so that they are symmetrically with respect to the length of the bed

2. Place the beam on the knife-edges on the blocks so as to project equally beyond each knife-edge. See that the load is applied at the center of the beam.

3. Note the initial reading of vernier scale.

4. Apply a load and again note the reading of the vernier scale.

5. Go on taking reading applying load in steps each time till you have minimum 6 readings.

6. Find the deflection (δ) in each time by subtracting the initial reading of vernier scale.

7. Draw a graph between load (W) and deflection (δ). On the graph choose any two convenient points and between these points find the corresponding values of W and δ. Putting these values in the relation $E = \frac{Wl^3}{48\delta I}$ Calculate the value of E.

8. Calculate the bending stresses for different loads using relation $\sigma_b = \frac{My}{I}$ as given in the observation table.

9. Repeat the experiment for different beams.

Result

a. Bending stress……..units
b. Young's modulus……units

Experiment No.7

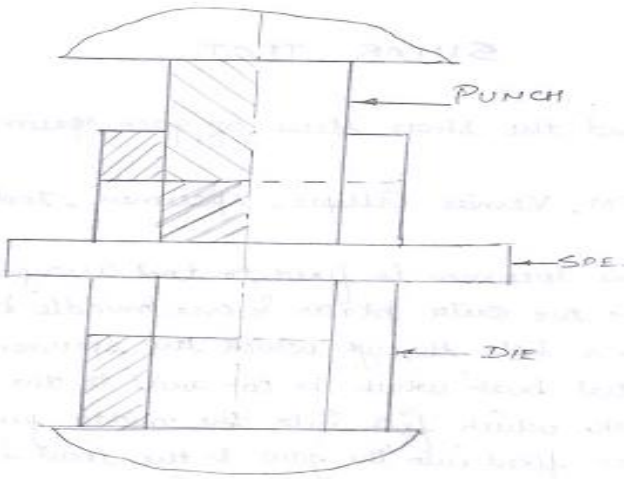
Title Shear test

Aim To find the shear strength of given specimen

Material and Equipment

1. Universal testing machine
2. Shear test attachment
3. Given specimen

Figure



Shearing fixture

Observation

Diameter of the pin d= ….mm
Cross sectional area of the pin(in double shear)

= 2 X π/4 Xd^2=…. mm^2

Load taken by the specimen at the time of failure, W =. ……(N)

Strength of the pin against shearing (τ)

$$\tau = \frac{4W}{2\pi d^2} = \ldots \text{ N/mm}^2$$

Procedure

1. Insert the specimen in position and grip one end of the attachment in the upper portion and one end in the lower position

2. Switch on the UTM

3. Bring the drag indicator in contact with the main indicator.

4. Select the suitable range of loads and space the corresponding weight in the pendulum and balance it if necessary with the help of small balancing weights

5. Operate (push) the button for driving the motor to drive the pump.

6. Gradually move the head control ever in left hand direction till the specimen shears.

7. Note down the load at which the specimen shears.

8. Stop the machine and remove the specimen.

Repeat the experiment with other specimens.

Precautions

1. The measuring range should not be changed at any stage during the test.

2. The inner diameter of the hole in the shear stress attachment should be slightly grater than the specimen.

3. Measure the diameter of the specimen accurately.

Result.

Shear strength of the specimen …….N/mm^2

Experiment No. 8

Title Compression test

Aim To find the compressive strength of given specimen.

Material and Equipment

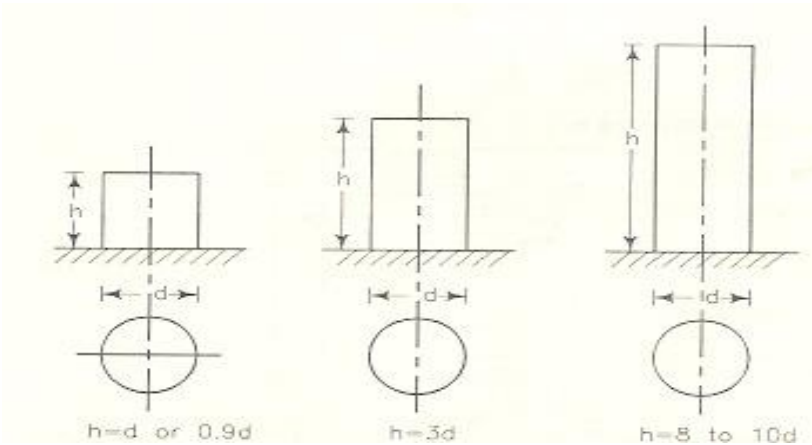
Universal testing machine,
Compression pads,
Given specimen,

Theory

This is the test to know strength of a material under compression. Generally compression test is carried out to know either simple compression characteristics of material or column action of structural members.
It has been observed that for varying height of member, keeping cross-sectional and the load applied constant, there is an increased tendency towards bending of a member.

Member under compression usually bends along minor axis, i.e, along least lateral dimension. According to column theory slenderness ratio has more functional value. If this ratio goes on increasing, axial compressive stress goes on decreasing and member buckles more and more. End conditions at the time of test have a pronounced effect on compressive strength of materials. Effective length must be taken according to end conditions assumed, at the time of the test. As the ends of the member is made plain and fit between two jaws of the machine, fixed end is assumed for calculation of effective length. Effective length is taken as 0.5 L where L is actual length of a specimen

Figure



Observation

Cross sectional area of the specimen perpendicular to the load=A=……mm^2

Load taken by the specimen at the time of failure, W=. ……(N)

Strength of the pin against shearing (σ) = [W/A] N/mm^2

Procedure

1. Place the specimen in position between the compression pads.
2. Switch on the UTM
3. Bring the drag indicator in contact with the main indicator.
4. Select the suitable range of loads and space the corresponding weight in the pendulum and balance it if necessary with the help of small balancing weights
5. Operate (push) the button for driving the motor to drive the pump.
6. Gradually move the head control ever in left hand direction till the specimen fails.
7. Note down the load at which the specimen shears
8. Stop the machine and remove the specimen.
9. Repeat the experiment with other specimens.

Precautions

1. Place the specimen at center of compression pads,
2. Stop the UTM as soon as the specimen fails.
3. Cross sectional area of specimen for compression test should be kept large as compared to the specimen for tension test: to obtain the proper degree of stability.

Result

Compressive strength of the specimen ……N/mm^2